COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	7/15/26	**DATE SUBMITTED:**	7/15/26
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ACTIVITY NO. _1__ LABORATORY ACTIVITY

1. PROCEDURE OUTPUT

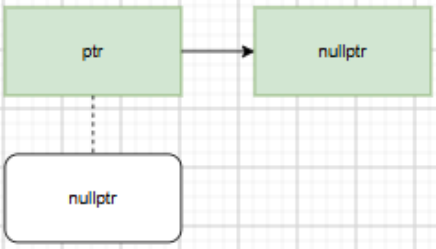
Draw the memory diagram for the following:

```
1. int *ptr = nullptr;
2. int value = 10;
   int  *ptr= &value;
3. int *ptr = new int;
4. int *ptr = new int(10);
```

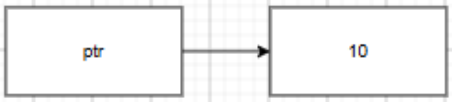
5. Differentiate Constructor, Destructor, Copy Constructor, Copy Assignment Operator.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

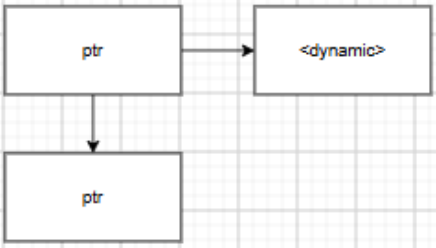
1. int *ptr = nullptr;



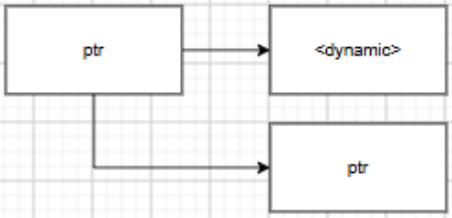
2. int value = 10;
int *ptr= &value;



3. int *ptr = new int;



4. in *ptr = int(10)



5. Differentiate Constructor, Destructor, Copy Constructor, Copy Assignment Operator.

Constructor	Destructor	Copy Constructor	Copy Assignment Operator
Initializes a new object	Cleans up an object before destruction	Creates a new object as a copy of another	Copies data into an existing object
`Class();`	`~Class();`	`Class(const Class &obj);`	`Class& operator=(const Class &obj);`
Object is created	Object goes out of scope or `delete` is called	New object initialized from another object	Existing object is assigned another object's value
Yes	No	Yes	No
None	None	None	Returns reference (`Class&`)
`A a;`	End of scope	`A b = a;`	`b = a;`

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT, Grammarly**

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Concept explanation, code assistance, grammar correction, spelling correction, formatting, and improving writing clarity.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **ChatGPT was used to explain C++ concepts, provide code guidance, and improve formatting, while Grammarly was used to correct grammar, spelling, and sentence structure. All work was reviewed, understood, and completed by me. The use of these tools did not replace my original thinking, analysis, or authorship.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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